a Data

3 technologies, 20 sections

DLPFC, 10x Visium (12)
STARmap mPFC (3)
MERFISH hypothalamus (5)

expert domain labels
(used only for scoring)

b Methods

13 published methods:
HiSTaR, GraphST, BANKSY,
SpaGCN, STAGATE, SEDR,
SpaceFlow, NichePCA, CCST,
Spatial-MGCN, DeepST,
BASS, BayesSpace
+ training-free: linear
diffusion, AnisoST, PCA

c Clustering protocols

Standard: one mclust-style fit
Robust: best likelihood of 20
Lottery: 20 single random starts
R mclust perturbations:
- permute spot order (10×)
- add 1% noise (10×)

same mixture model (EEE)

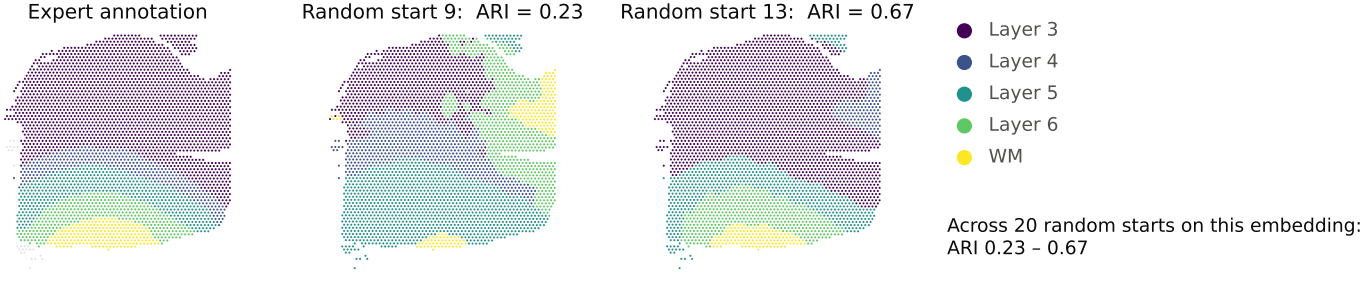
d Evaluation

ARI vs. labels
seed variance
runtime

Wilcoxon tests,
Holm correction

matched vs. standard

e Same fixed HiSTaR embedding (DLPFC 151669), same mixture model, different random start only



Across 20 random starts on this embedding:
ARI 0.23 – 0.67